Deep Neural Networks & Regularization – The Battle Against Overfitting

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1. Introduction

This project implements a deep neural network using TensorFlow/Keras on the CIFAR-10 dataset, focusing on combating overfitting through regularization techniques. The project evaluates model performance with and without regularization and applies methods such as L2 regularization, dropout, early stopping, and data augmentation to improve generalization.

2. Objectives

- Train a fully connected neural network (MLP) on the CIFAR-10 dataset.
- Apply and compare various regularization techniques (L2, Dropout, Early Stopping).
- Evaluate the impact of regularization on training stability and generalization.
- Justify the choice of hyperparameters.

3. Model Architecture

The baseline model consists of three dense layers: 512, 256, and 128 units respectively, using ReLU activation functions. A regularized model was also developed using L2 regularization (0.0001), dropout (rate of 0.3), and early stopping. Data augmentation techniques such as horizontal flipping, zoom, and rotation were applied to enhance generalization.

4. Experimental Setup

The CIFAR-10 dataset was normalized by scaling pixel values to the range [0, 1], and the labels were one-hot encoded. A batch size of 64 was used. Both baseline and regularized models were trained using the Adam optimizer with a learning rate of 0.0001.

5.Justification of Hyperparameter Choices

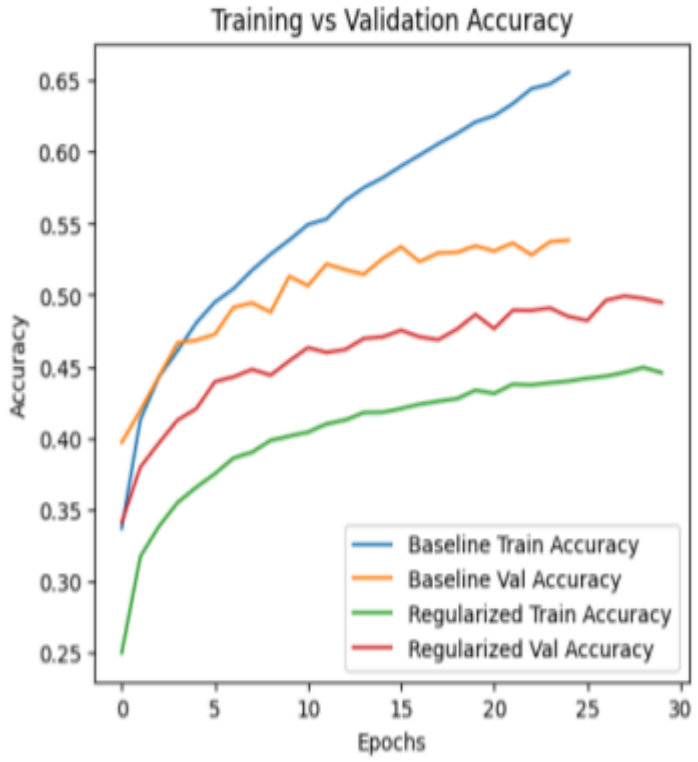
A low learning rate of 0.0001 was selected to ensure stable training. A batch size of 64 balanced computational efficiency and gradient quality. The model used two hidden layers (512 and 256 units) with ReLU activation to provide sufficient capacity without overfitting. A dropout rate of 0.3 reduced neuron co-adaptation, and L2 regularization with a weight decay of 0.0001 helped prevent weight explosion. Early stopping with patience=7 was applied to improve generalization and avoid overfitting.

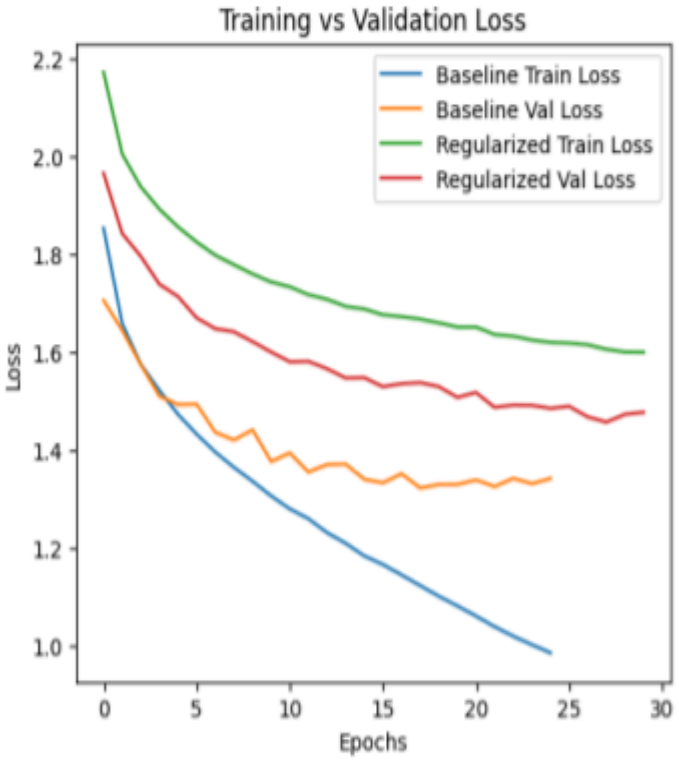
6. Training Stability & Convergence Analysis

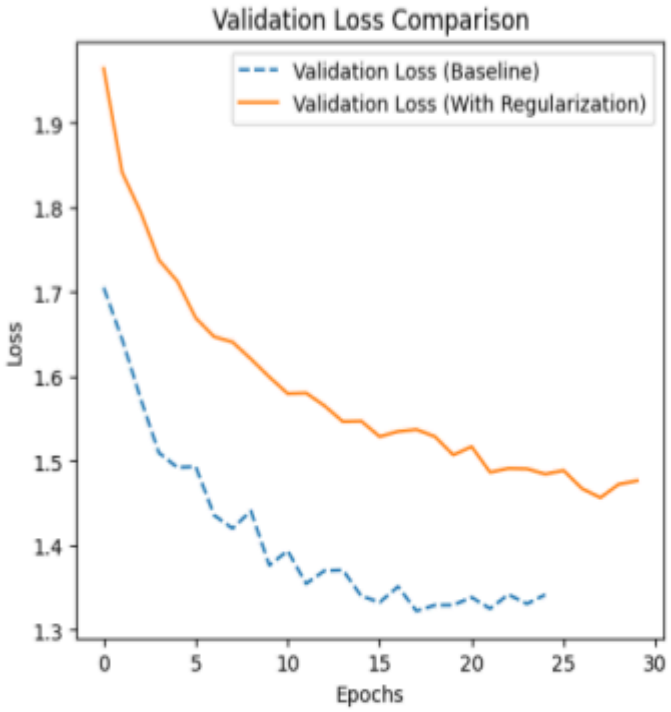
The baseline model, trained without any regularization, showed signs of overfitting: training loss decreased steadily while validation loss began to rise, indicating poor generalization. In contrast, the regularized model, which incorporated L2 regularization, dropout, and early stopping, exhibited more consistent training and validation loss curves. This indicates improved stability and convergence, with early stopping ensuring that the model retained optimal weights and avoided unnecessary over-training.

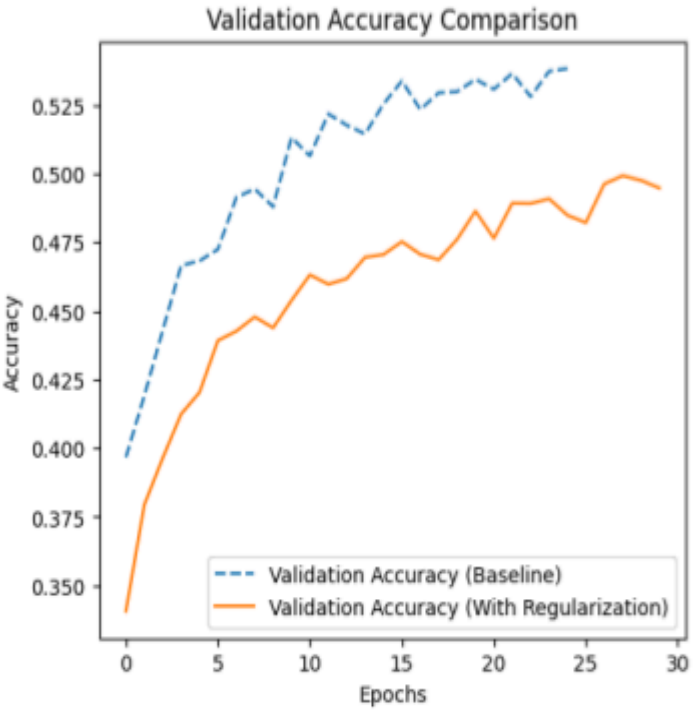
7. Results

The regularized model achieved better validation accuracy and smoother loss curves compared to the baseline model. Regularization techniques helped reduce overfitting and improved the model's ability to generalize to unseen data










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Evaluation completed and results plotted.
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8. Conclusion

This project demonstrated the importance of regularization techniques in deep learning. L2 regularization, dropout, early stopping, and data augmentation collectively contributed to enhanced training stability and generalization. The practical application of these methods proved effective in reducing overfitting and improving model performance on the CIFAR-10 dataset.

9. References

- TensorFlow Documentation: https://www.tensorflow.org/

- Keras API Reference: https://keras.io/api/

- CIFAR-10 Dataset: https://www.cs.toronto.edu/~kriz/cifar.html